

TASTY, Kazakhstan

WMO: 380810

Lat: 44.800N

Lon: 69.117E

Elev: 190

StdP: 99.06

Time Zone: 6.00 (E06)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.9	-20.5	-26.6	0.3	-22.1	-24.3	0.4	-20.0	14.0	-7.2	10.5	-6.6	2.8	90	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	38.3	18.3	36.6	17.7	35.1	17.3	19.6	33.3	18.8	33.0	18.1	32.3	3.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	11.3	22.2	13.9	10.1	23.0	12.2	9.1	23.2	56.7	33.4	54.2	33.3	51.8	32.5	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	10.3	9.4	-25.5	42.1	4.3	2.3	-28.6	43.7	-31.1	45.0	-33.5	46.3	-36.6	47.9		
(5)			-25.7	21.7	4.2	1.5	-28.7	22.8	-31.2	23.7	-33.6	24.6	-36.6	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-7.6	-7.0	0.6	12.3	19.1	25.0	27.5	24.7	17.9	9.2	0.8	-5.5		
	DBStd	13.61	6.18	6.39	6.15	4.96	4.50	3.73	3.03	3.07	4.17	5.13	5.74	6.78		
	HDD10.0	2189	547	477	296	33	3	0	0	0	1	76	277	481		
	HDD18.3	3907	805	710	550	188	43	2	0	1	58	285	524	739		
	CDD10.0	2128	0	0	3	102	284	449	542	456	238	51	2	0		
	CDD18.3	803	0	0	0	7	66	202	283	198	44	3	0	0		
	CDH23.3	11145	0	0	2	104	793	2875	4103	2580	636	52	0	0		
	CDH26.7	5893	0	0	0	24	297	1583	2414	1327	238	10	0	0		
	WSAvg	3.4	3.3	3.9	4.1	4.3	3.6	3.1	2.9	3.1	3.2	3.2	3.3	3.0		
(15) Precipitation	PrecAvg	158	17	15	16	19	20	8	12	4	2	9	21	15		
	PrecMax	285	27	46	38	46	60	39	56	27	10	35	77	46		
	PrecMin	91	6	1	1	0	3	0	0	0	0	1	2			
	PrecStd	57	6	10	11	13	17	10	15	6	3	11	23	10		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	9.0	20.0	29.8	35.0	39.2	41.4	39.1	34.3	27.8	17.3	9.5		
		MCWB	4.9	4.5	9.7	15.6	16.4	18.4	19.0	18.6	16.6	14.4	10.4	5.7		
	2%	DB	4.1	4.9	15.3	26.4	32.1	37.4	39.0	36.8	31.7	24.4	13.9	5.5		
		MCWB	2.0	2.3	8.5	13.7	15.9	17.7	18.7	17.6	15.3	12.8	9.0	3.4		
	5%	DB	1.9	2.4	11.8	23.7	30.0	35.9	37.2	34.9	29.5	21.2	11.0	3.6		
		MCWB	0.6	0.4	7.0	12.1	15.0	17.4	18.2	17.0	14.5	11.2	6.8	2.0		
	10%	DB	0.4	0.8	8.6	21.2	27.8	34.3	35.5	33.0	27.2	18.2	8.4	2.1		
		MCWB	-0.6	-0.6	4.9	11.2	14.4	16.8	17.7	16.4	13.5	9.6	5.3	0.9		
	0.4%	WB	5.2	4.7	12.6	16.8	18.5	19.7	21.5	19.7	17.9	15.2	11.2	5.7		
		MCDB	7.4	8.6	18.4	26.0	27.7	34.7	34.1	32.6	31.8	25.8	15.0	8.5		
	2%	WB	2.3	2.3	9.1	15.0	17.3	18.7	20.1	18.4	15.9	13.2	9.3	3.6		
		MCDB	3.7	4.2	13.4	23.0	26.6	34.0	33.4	33.4	29.6	22.7	13.4	5.3		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	0.6	0.6	7.1	13.4	16.4	18.0	19.3	17.6	14.8	11.6	7.4	2.3		
		MCDB	1.6	2.1	12.0	21.9	26.3	33.0	32.9	32.0	28.5	20.1	10.3	3.4		
	10%	WB	-0.5	-0.5	5.0	12.0	15.5	17.4	18.7	17.0	13.7	10.0	5.7	0.9		
		MCDB	0.4	0.8	8.8	19.6	25.4	32.1	32.6	30.9	26.3	17.5	8.1	2.0		
		MDBR	7.0	8.2	8.8	12.3	13.3	14.9	14.6	14.8	15.0	13.0	8.5	6.7		
		MCDBR	7.4	8.6	13.5	15.9	16.3	16.7	16.5	17.4	18.0	16.9	12.5	7.3		
(36) Mean Daily Temperature Range	5% DB	MCWBR	5.8	6.7	7.8	7.0	6.0	5.9	6.0	6.5	8.1	8.5	8.1	5.8		
		MCDBR	6.9	8.1	12.4	13.6	13.4	15.1	14.6	15.1	17.0	15.9	10.8	7.1		
	5% WB	MCWBR	5.5	6.5	7.5	7.3	5.4	5.7	5.5	6.0	8.0	8.3	7.5	5.7		
(40) Clear-Sky Solar Irradiance	taub		0.352	0.384	0.434	0.470	0.471	0.450	0.439	0.406	0.376	0.390	0.382	0.352		
	taud		2.019	1.989	1.982	1.945	1.988	2.090	2.141	2.216	2.253	2.131	2.095	2.065		
	Ebn at Noon		709	764	782	791	805	823	828	843	838	753	672	663		
	Edh at Noon		108	134	156	176	174	158	148	132	117	115	100	94		
(44) All-Sky Solar Radiation	RadAvg		1.50	2.52	3.91	5.47	6.97	7.60	7.37	6.68	5.23	3.31	1.85	1.23		
	RadStd		0.25	0.29	0.31	0.45	0.37	0.32	0.35	0.23	0.20	0.25	0.19	0.17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page